

NADINE EL-MUFTI

Canadian Citizen, residing in  Montreal, QC

 (514) 802-3932 |  |  |  |

EXECUTIVE SUMMARY

Clinically grounded AI researcher specializing in speech biomarkers for early disease detection. Master's candidate at Concordia & Mila with end-to-end ML experience and a strong focus on ethics, fairness, and human-centered healthcare technology.

WORK EXPERIENCE

The Douglas Research Centre, McGill University

Graduate Research Assistant

Montreal, QC

Jun 2025 – Present

- Driving speech-based relapse prediction research in psychotic disorders in collaboration with Dr. Lena Palaniyappan and Dr. Alban Voppel by coordinating multisite data collection using REDCap workflows, and powering the Québec Speech Bank by engineering and analyzing pipelines to clean, align, and structure longitudinal speech data across diverse studies. This work enables multimodal insights into voice, symptoms, and cognition, paving the way for more predictive, compassionate mental health care.

Applied Perception Lab, Concordia University

Graduate Research Assistant

Montreal, QC

Dec 2024 – Present

- Driving research at the intersection of AI, ethics, and equity, diversity, and inclusion (EDI) to develop actionable frameworks that address pressing societal challenges and advance responsible, human-centered healthcare technology.

Mila - Quebec AI Institute

Collaborating Researcher

Montreal, QC

Oct 2024 – Present

- Leading novel research in Medical Machine Learning for early disease detection via speech analysis, covering the entire ML lifecycle from dataset curation to model training, evaluation, optimization, and deployment.

Concordia University

Teaching Assistant

Montreal, QC

Jan 2025 – April 2025

- Achieving consistent and fair grading for Concordia's Machine Learning course 'Comp 432' by marking labs, delivering prompt feedback, and addressing inquiries from 110 students.

Applied Perception Lab

NSERC USRA Research Assistant

Montreal, QC

May 2022 – Aug 2022

- Co-authored a conference paper for Medical Image Computing and Computer Assisted Intervention (MICCAI) 2022's Ethical and Philosophical Issues in Medical Imaging (EPIMI) workshop, presenting a comprehensive User-Centered Design approach to identify and address Ventriculostomy surgeons' workflow challenges, to improve surgical efficiency.

PUBLISHED & ON-GOING RESEARCH

- Automatic Speech Recognition Performance in Psychiatric Speech: Linguistic, Clinical, and Architectural Factors (*In Preparation*)
- Beyond the Scalpel: Envisioning the Societal and Ethical Dimensions of Future Surgical Technologies (*In Preparation*)

- Enhancing Speech-Based Parkinson's Detection with Celebs4PD: A Corpus of Natural, In-the-Wild, Longitudinal Celebrity Speech (*In Preparation*)
- [User-Centered Design for Surgical Innovations: A Ventriculostomy Case Study](#)
Lecture Notes in Computer Science — Volume 13755 · Dec 20, 2022

LEADERSHIP, SERVICE & ENGAGEMENT

[The Levantines](#) (Member, Feb 2026) · [Association Étudiante de Mila AÉM](#) (Member, Oct 2025) · [Arbour Foundation Scholars](#) (Member, Sep 2025) · [Golden Key International Honour Society](#) (Member, Sep 2025) · [Brainhack MTL](#) | [Brainhack Global](#) (Organizer, Jul 2025) · [Graduate Students' Association of Concordia GSA](#) (Subcommittee Member, Jan 2025) · [School of Graduate Studies, Concordia University](#) (Ambassador, Sep 2024) · [GEMinAI](#) | [Concordia AI2](#) (Mentee, Oct 2024 – May 2025)

EDUCATION

Master of Computer Science Computer Science <i>Concordia University / Mila - Quebec AI Institute</i>	Montreal, Quebec Jan 2025 – Present
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Bachelor of Engineering Software Engineering GPA (3.67) With Distinction <i>Concordia University</i>	Montreal, Quebec Jan 2019 – May 2024
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HONORS & AWARDS

Arbour Foundation Master's Scholarship Issued by The Arbour Foundation	Sep 2025
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FRQNT B1X Award Competition – 92/100, Ranked 9th (Funding Cutoff: 8) Issued by the Fonds de recherche du Québec	Apr 2025
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NSERC CGS-M Award Issued by the Natural Sciences and Engineering Research Council of Canada	Apr 2024
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NSERC Undergraduate Student Research Award (USRA) Issued by the Natural Sciences and Engineering Research Council of Canada	May 2022
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LICENSES & CERTIFICATIONS

[Bell x Mila x Buzz HPC x Kids Help Phone Hackathon](#) (*Mila, Apr 2026*) · [Ethics and Governance of AI for Health](#) (WHO Academy, 2026) · [QcSE Program](#) (V1 Studio, 2025) · [Strategic Public Communications](#) (GradProSkills, 2025) · [TRAIL AI Learning Certificate](#) (Mila, 2025) · [TCPS 2: CORE 2022](#) (Government of Canada, 2025) · [Tech Stewardship Certification](#) (Tech Stewardship Network, 2024)